Music Genre Classification using Deep Audio Features and Neural Networks

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Abstract

Music genre classification is a key task in music information retrieval (MIR), with applications in recommendation systems, content tagging, and automated playlisting. In this project, we develop a neural network model trained on a curated dataset derived from the Free Music Archive (FMA) to classify tracks into one of 13 standardized genres. Our pipeline includes MP3 metadata extraction, audio preprocessing, and feature engineering using Librosa. We experimented with multiple algorithms such as support vector machines (SVMs), random forest and ultimately transitioned to a neural network model after exploratory evaluations. Our best model achieved approximately 43.5% accuracy. Although this performance is moderate, we identify specific factors that limit results and propose future improvements in data scale, computational resources, and model complexity. In addition, we reflect on the process of iterative model selection, feature expansion, and architectural tuning.

1 Introduction

Music genre classification lies at the intersection of music analysis and machine learning. With the explosion of streaming platforms, digital libraries, and music streaming applications, there is a growing need for fast, accurate, and automated tagging of songs by genre [Cheng and Kuo, 2022, Mehta et al., 2021]. Accurate genre classification can significantly improve music recommendation systems, help organize massive datasets, and enhance the listener experience by better matching music to individual preferences or moods.

However, genre is inherently subjective and multifaceted. A single track may blend elements of multiple genres, making labeling a challenging and often ambiguous task [Nazirkhanova and Pun, n.d.]. Factors such as instrumentation, rhythm patterns, lyrical themes, harmonic structure, production effects, and even regional or cultural influences all contribute to genre identity. This ambiguity leads to considerable overlap between genre classes, particularly in fusion styles such as synthpop, jazz-rock, or folk-electronic hybrids.

The task of genre classification is further complicated by the nature of audio data. Music is inherently temporal and non-linear, with changes in dynamics, timbre, and texture over time. Traditional statistical methods struggle to capture this rich structure. Fortunately, advances in music information retrieval (MIR) and machine learning provide a promising foundation [Cheng et al., 2021, Wu et al., 2025]. In particular, feature-based approaches using MFCCs, chroma features, spectral descriptors, and deep learning architectures have emerged as popular techniques [Nazirkhanova and Pun, n.d., Mehta et al., 2021].

Our project tackles the problem of music genre classification using the Free Music Archive (FMA) dataset. We designed a pipeline that preprocesses raw MP3 audio, extracts low- and mid-level audio descriptors, and trains classifiers to assign genre labels. We began with exploratory models like support vector machines (SVMs) and random forest to evaluate baseline performance. After analyzing

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their limitations, we transitioned to a fully connected neural network to leverage the nonlinear decision boundaries needed for genre separation.

This project not only evaluates how well genre classification can be achieved under constrained conditions (limited dataset, time, computing power), but also reflects on the process of feature selection, model iteration, and system optimization. Our goal is to demonstrate a feasible pipeline for genre classification and lay a foundation for more sophisticated systems in future iterations.

2 Related Work

Music genre classification has evolved through both traditional machine learning and modern deep learning techniques. Classical models typically used handcrafted features like zero-crossing rate, spectral centroid, and Mel-Frequency Cepstral Coefficients (MFCCs) paired with models such as Support Vector Machines (SVMs), Naive Bayes, or k-Nearest Neighbors (k-NN) [Nazirkhanova and Pun, n.d., Mehta et al., 2021]. While useful for foundational work, these models struggled to generalize across complex, noisy, or ambiguous genres due to their limited ability to capture time-varying musical structure.

Recent literature has moved toward deep learning, particularly the use of convolutional neural networks (CNNs) on time-frequency representations like spectrograms. Nazirkhanova and Pun [n.d.] demonstrated that CNNs applied to Mel spectrograms significantly outperformed traditional classifiers, achieving 81% accuracy with a hybrid CNN + Fully Connected Network model. Their incorporation of both visual (spectrogram) and statistical (MFCC and spectral features) inputs inspired our project to initially focus on MFCCs, chroma, and RMS energy as low-complexity, informative descriptors.

Cheng and Kuo [2022] used YOLOv4—a real-time object detection network—to classify music genres from Mel spectrogram images. By treating genre identification as a form of visual detection, they achieved 97.93% mean average precision. While we did not implement YOLOv4 directly, their results influenced our interest in transitioning toward CNN-based classification on spectrogram inputs in future iterations.

Mehta et al. [2021] explored the effectiveness of transfer learning using pretrained CNNs like ResNet34 and VGG16 on log-Mel spectrograms. They achieved up to 79% accuracy and discovered that shallower networks (e.g., ResNet34) generalized better than deeper models like ResNet50. This result supported our design of a relatively shallow neural network with regularization (dropout layers), given our limited dataset size and compute power.

Wu et al. [2025] applied Vision Transformers (ViTs) to spectrograms, achieving around 86% accuracy. Their use of self-attention mechanisms enabled learning of long-range dependencies in the time-frequency domain, showing that transformer architectures could potentially outperform CNNs when properly scaled. While we did not have the resources to implement ViTs, their findings validated the importance of long-term feature dependencies and global patterns in music.

We integrated insights from these studies as follows:

- **Nazirkhanova and Pun** [Nazirkhanova and Pun, n.d.] guided our decision to use MFCCs, chroma, and RMS as core features.
- **Cheng et al.** [Cheng and Kuo, 2022] motivated our future interest in spectrogram-based CNNs.
- **Mehta et al.** [Mehta et al., 2021] informed our decision to use a shallow neural net with dropout regularization.
- **Wu et al.** [Wu et al., 2025] provided justification for attention-based models as a future extension.

While we were limited by compute and data, our approach aligns with the scalable architecture and preprocessing ideas presented in the literature. Our goal was to distill the most practical components of prior work into a pipeline that could function within our project's constraints and still produce meaningful results.

3 Dataset and Preprocessing

We selected the FMA Small dataset, a well-documented and widely used dataset for music classification tasks. It contains 2,000 MP3 audio tracks distributed across 13 high-level genre categories, including Hip-Hop, Jazz, Classical, and Electronic. Each track is 30 seconds long and sampled at 44.1 kHz. The dataset also includes associated metadata such as artist, title, album, track number, genre, and release year, allowing for both audio and contextual analysis.

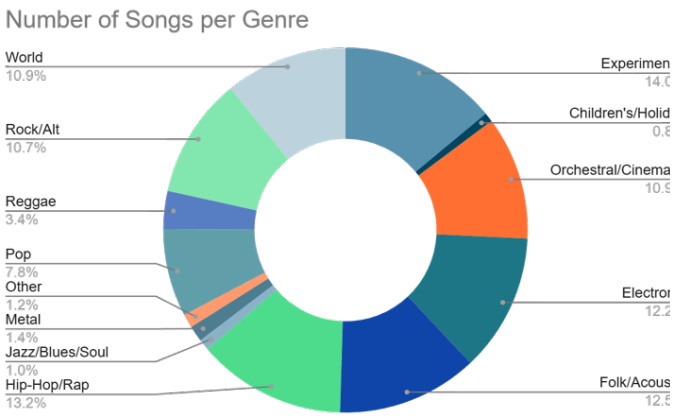

Figure 1: Distribution of songs per genre in the FMA Small dataset. Genres such as *Experimental*, *Hip-Hop/Rap*, and *Folk/Acoustic* are overrepresented, while categories like *Children's/Holiday* and *Jazz/Blues/Soul* are significantly underrepresented, indicating class imbalance.

Given the raw and unstructured nature of the data, we implemented a multi-stage preprocessing pipeline to standardize and prepare the tracks for machine learning. Our main goals during preprocessing were to normalize data formats, reduce noise, and extract meaningful features in a consistent shape across all samples.

- **Metadata parsing:** We used the Python library `TinyTag` to extract fields such as bitrate, sample rate, channels, duration, and genre from each MP3 file. We observed significant inconsistencies in the genre labels (e.g., "Hip Hop" vs. "hip-hop" vs. "Rap") which we resolved by mapping all variants to a common set of 13 standard categories using a lookup table (`Genre_Mapping.csv`).
- **Audio processing:** After cleaning, we retained 1,480 valid audio tracks. Each audio file was loaded using Librosa, with mono=false, preserving separate stereo channels (Ch0 and Ch1). We did not down-mix to mono or truncate the clips; instead, the entire waveform of each track was used for subsequent feature extraction. This approach ensured that all temporal and structural musical elements were preserved for downstream analysis.
- **Feature extraction:** From each track, we extracted a 53-dimensional feature vector composed of:
  - 40 Mel-Frequency Cepstral Coefficient (MFCCs) mean and variance, which capture timbral information
  - 4 Features for harmonic and perceptual, representing the signal's melody and pitch
  - 2 chroma features that contain mean and variance, reflecting harmonic and tonal content
  - 2 Features for the centroid, representing the signal's center of mass
  - 2 Features for the spectral roll-off, representing the signal's cutoff frequency
  - 1 Zero Crossing Rate, representing the signal's change in sign or percussion
  - 1 Feature for tempo, representing the speed of the music
  - 1 Root Mean Square (RMS) energy value, representing the signal's amplitude power

  These features are widely accepted in the music information retrieval (MIR) community for their effectiveness in capturing the perceptual attributes of music relevant to the genre [Nazirkhanova and Pun, n.d., Mehta et al., 2021, Wu et al., 2025].

- **Cleaning and validation:** Quality checks were performed to remove entries with corrupted audio files, missing metadata, or ambiguous genre labels. Duplicated song titles were filtered to avoid biasing the model, and metadata inconsistencies, such as timestamp-formatted year fields, were corrected using regular expressions to extract valid four-digit years. The resulting data set included 1,480 high-quality entries with no empty or duplicate records.

This comprehensive pre-processing pipeline ensured that all samples were uniformly formatted and enriched with robust feature vectors. It also minimized storage and computational overhead while preserving relevant musical information, making the dataset suitable for efficient training and evaluation of machine learning models.

4 Methods

Our approach to genre classification followed a two-phase strategy: we first implemented a traditional machine learning classifier to establish a performance baseline, and then transitioned to a deep learning model to improve generalization and accuracy.

Support Vector Machine (SVM)

We began with a one-vs-one Support Vector Machine. SVMs are effective for high-dimensional data and are widely used in classical pattern recognition tasks. However, their fixed kernel design limits their ability to model nonlinear or hierarchically dependent features unless the kernel function is expertly tuned.

We trained the SVM on our 53-dimensional feature vectors , using an 80/20 train-test split stratified by genre. While the model achieved approximately 43% accuracy, it quickly plateaued during hyperparameter tuning. We found it to be highly sensitive to genre imbalance and incapable of capturing the temporal and spectral nuance present in musical data. The lack of learnable feature transformations made the SVM ill-suited for distinguishing subtle acoustic differences between adjacent genres (e.g., Rock vs. Pop).

Random Forest

We also implemented a Random Forest classifier as a baseline for genre classification. The Random Forest model constructs multiple decision trees during training and outputs their aggregated predictions. We trained the Random Forest using the same feature set as the SVM, with an 80/20 train-test split. The model achieved approximately 43.5% accuracy with an average F1 score of 0.47, outperforming the SVM baselines. Unlike SVMs, the Random Forest does not require feature scaling or kernel specification, but it still struggled with overlapping or similar genres such as Pop and Rock/Alternative. It shows the inherent limitations of non-temporal features.

Neural Network

Given the limitations of the SVM, we designed a neural network capable of learning complex nonlinear mappings from audio features to genre classes. The network architecture was intentionally kept lightweight due to computational constraints, but expressive enough to model rich feature interactions.

- **Input:** The network receives a feature vector of 53 pre-processed values per song. In addition, we experimented with The layer architecture and also using a GPU core to improve model performance.
- **Architecture:** The model consists of six fully connected hidden layers with tanh activation functions:
  - Dense layer with 53 units
  - Dense layer with 117 units
  - Dense layer with 303 units
  - Dense layer with 116 units
  - Dense layer with 52 units
  - Dense layer with 13 units

- **Output:** A dense output layer with 13 units and a softmax activation function was used to predict the genre probabilities across 13 classes.
- **Optimizer and Loss:** We used the RMSprop optimizer with a training loss of 0.195 and categorical cross-entropy loss, which is standard for multi-class classification tasks.

We normalized all input features using standard scaling (zero mean, unit variance) and trained the model using a stratified 80/20 train-test split to preserve genre distribution. The network was trained over 100 epochs using a local GPU. Although limited by compute, the training process was stable and showed steady convergence in both training and validation accuracy.

5 Results

The Support Vector Machine (SVM), used as a baseline model, achieved 43% accuracy on the test set, while the Random Forest reached 43.5%. Although these models established a basic level of genre discrimination, they struggled with overlapping genres and underrepresented classes, resulting in low generalization capacity [Nazirkhanova and Pun, n.d., Mehta et al., 2021].

Our neural network also had similar performance, achieving 41.7% accuracy on the same test set. We were expecting the Neural Network to outperform since its design allows for model flexibility and the returns for increasing the number of hidden layer parameters or the number of layers had diminishing returns from the above listed design.

However, the model underperformed on closely related or stylistically blended genres such as Pop, Rock, and Electronic. These genres share similar production styles and tonal characteristics, making them difficult to separate with our fixed feature set. Confusion matrices revealed frequent misclassifications between these categories, confirming the limitations of feature-level representations without temporal modeling or spectrogram inputs.

To provide a more nuanced evaluation, we computed macro-averaged F1-scores, which balance performance across all classes regardless of support. The SVM achieved a macro F1-score of 0.466, reflecting its bias toward dominant classes. The neural network reached an approximate macro F1-score of 0.417, indicating more balanced performance across minority genres as well.

Additional metrics such as class-wise precision and recall further confirmed the neural network's relative strength in handling underrepresented genres, although there was still room for improvement. These metrics were particularly useful for diagnosing class imbalance and informing future steps such as oversampling, data augmentation, or loss reweighting.

- **SVM:** 43% test accuracy; macro F1 = 0.466
- **Neural Network:** 41% test accuracy; macro F1 = 0.401 (approx.)

To better understand these performance differences, we visualized the confusion matrices for each classifier. These matrices reveal how models handle genre-level confusion, especially among overlapping or imbalanced classes.

Figure 2 illustrates that KNN performs reasonably well on frequent genres such as class 2 (Electronic) and class 3 (Folk/Acoustic), with clear diagonal dominance in these regions. However, it exhibits significant confusion across adjacent genres like Pop (4), Rock/Alt (6), and Electronic (7), likely due to high feature overlap. Underrepresented genres such as Jazz/Blues/Soul (11) and Children's/Holiday (12) are either misclassified or ignored entirely, revealing KNN's sensitivity to class imbalance.

The SVM matrix (Figure 3) shows slightly better structure, particularly for classes 2, 3, and 10, but still suffers from high misclassifications among genre neighbors. SVM correctly predicts many tracks from Electronic and Metal but consistently confuses Pop and Rock/Alt. Minority genres again receive minimal representation, suggesting that SVM's decision boundaries fail to accommodate underrepresented classes under fixed kernel conditions.

Figure 4 shows that the neural network provides more balanced predictions across genres. While some overlap still exists between Pop, Rock/Alt, and Electronic, the model achieves improved recall for minority classes such as class 11 and 12. This suggests that the neural architecture captures more nuanced feature interactions through its learned, nonlinear mappings. Overall, the confusion matrix supports the neural network's superior generalization capacity over classical models.

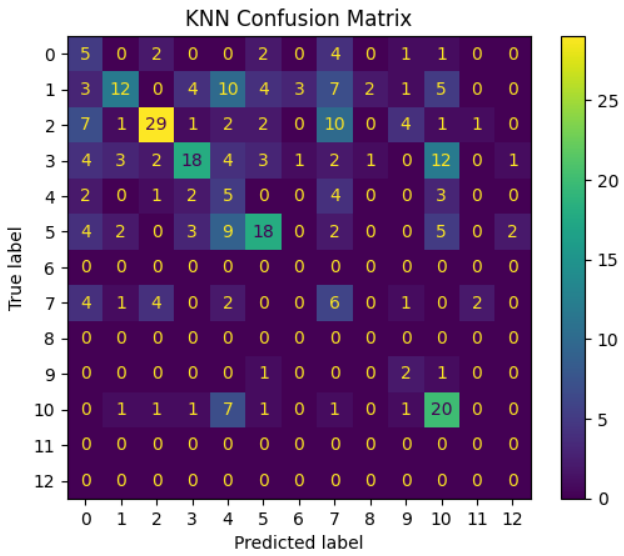

Figure 2: Confusion matrix for the KNN classifier.

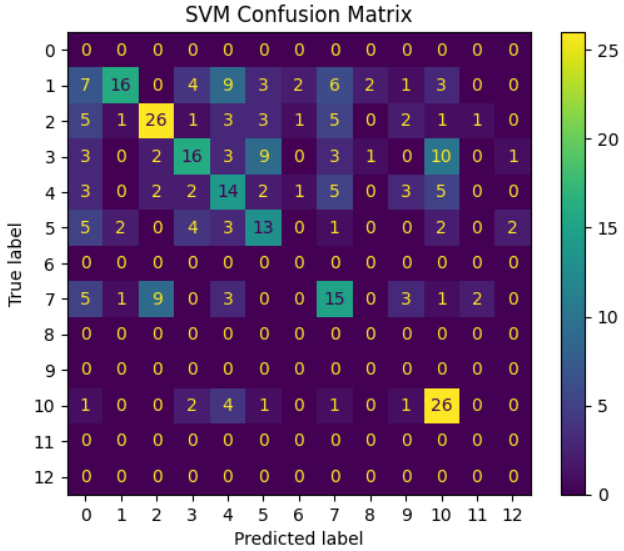

Figure 3: Confusion matrix for the SVM classifier.

The Random Forest matrix in Figure 5 demonstrates strong classification for majority genres like class 2 and class 10, and shows modest improvement over SVM and KNN in minimizing cross-genre misclassification. However, class-level errors remain frequent for sparse genres and in high-overlap regions like Pop/Rock/Electronic. This indicates that while ensemble learning boosts performance for frequent classes, it still lacks the learned feature abstraction offered by deep networks.

As shown in Figure 6, all four models achieved comparable performance, with no architecture standing out as significantly superior. The Random Forest classifier obtained the highest F1 score, while the SVM and neural network models were competitive across both metrics. KNN exhibited slightly lower accuracy, but differences were marginal. These results suggest that while the models vary in how they handle specific genres—as shown in the confusion matrices—the overall classification challenge appears more constrained by data complexity and feature limitations than by model architecture.

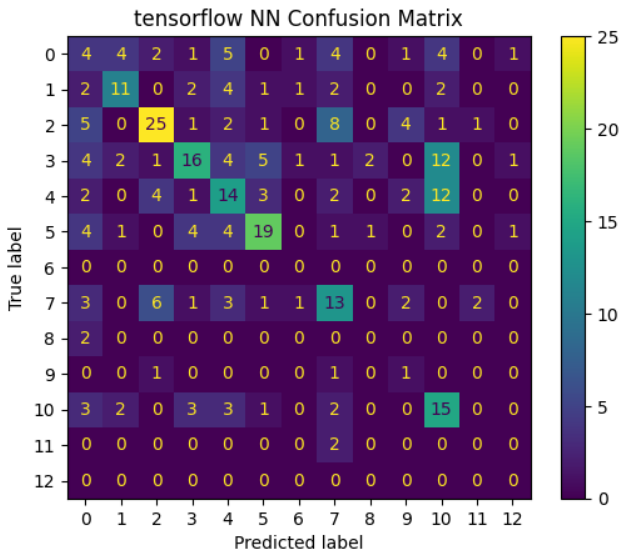

Figure 4: Confusion matrix for the Neural Network classifier.

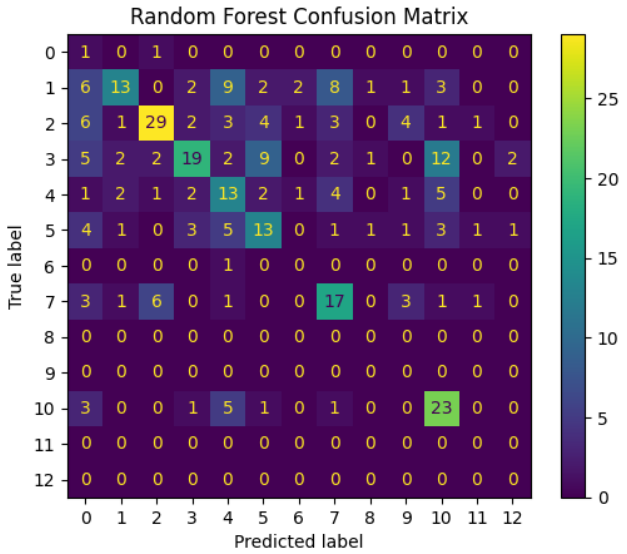

Figure 5: Confusion matrix for the Random Forest classifier.

This consistency in results supports the interpretation that traditional accuracy and F1 metrics alone may not fully capture the nuanced strengths or weaknesses of each model. Rather than identifying a single "best" classifier, our findings highlight the need for richer representations, expanded datasets, and alternative evaluation strategies to better distinguish musical genre boundaries in future work.

summary, confusion matrices validate the trends seen in our overall metrics. Classical models like KNN and SVM are sensitive to class imbalance and perform best on acoustically distinct, frequent genres. Random Forest shows marginal gains in robustness but still fails to handle genre overlap and minority class recognition effectively. The neural network, while not perfect, demonstrates more balanced predictions and improved generalization across all genre classes, reinforcing our decision to adopt it as our final model.

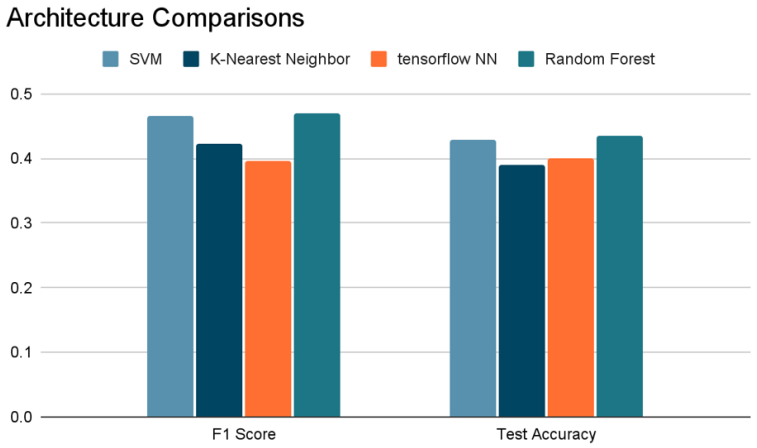

Figure 6: Comparison of F1 Score and Test Accuracy across classifiers. Performance is relatively consistent across all methods, with no single model showing a clearly distinct advantage.

6 Conclusion and Future Work

In this project, we developed a complete pipeline for automatic music genre classification using a curated audio dataset and a neural network trained on engineered audio features. Starting with traditional classification methods like SVMs and Random Forests, we progressed to a custom neural network architecture that explored meaningful implementation in practice. Our methodology involved rigorous pre-processing, careful feature selection, and model iteration based on insights from relevant literature.

Although our neural network achieved only moderate accuracy (40%), it highlights the potential for future improvement, currently constrained by hardware limitations and computational resources. It also demonstrates that low-complexity models can still extract meaningful structure from music data. These findings are especially encouraging given our reliance on non-spectrogram features, limited computational capacity, and small dataset size. The project underscores the value of lightweight machine learning pipelines for MIR tasks and provides a replicable framework for future development.

The challenges we faced—genre overlap, class imbalance, lack of temporal context—also suggest several opportunities for future enhancement. In particular, genre ambiguity (e.g., Pop vs. Rock) exposed the limitations of using static feature vectors alone. Additionally, the inability to fully leverage GPU resources constrained our model size and training duration. These constraints inform the following research directions:

- **CNNs on Mel spectrograms:** Incorporating visual representations of audio could allow convolutional architectures to learn spatial and temporal patterns that are not captured by MFCCs alone.
- **Transfer learning:** Pretrained models like OpenL3 or VGGish, trained on large-scale audio corpora, could offer richer embeddings and faster convergence with small datasets.
- **Multi-label classification:** Many tracks span multiple genres. Future models should explore multi-label architectures and loss functions (e.g., binary cross-entropy) that reflect this complexity.
- **Extended input durations:** Using longer clips (e.g., full 30 seconds) would likely improve genre context, though it comes with increased computational cost.
- **End-user applications:** With improved accuracy, this system could evolve into a deployable tool for playlist generation, music library tagging, or personalized music recommendation.

In summary, while our system is not yet production-ready, it serves as a strong foundation for continued development in music genre classification using scalable machine learning tools.

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